

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have worked with the Kisumu tray of eight substances (A table salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar) over ten lessons. Use your own tray evidence and your learning from the lessons to answer the Driving Question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? Answer each section in the space under its prompt. Use the letters A-H and the exact names of the substances. Use numbers and units (g, mL, C, mg) from your own results where you can. Write in full sentences and show your working for any calculation. Remember our class value of respect: write in your own words and be ready to listen to other groups' ideas.

Section 1: Same look, different behaviour

Look at the data table from the Kisumu tray tests (1 level spatula, about 2 g, of each solid in 50 mL of water at about 25 C; heat test done by the teacher).

Substance	In water	Heat test
A table salt	dissolves, clear solution	crackles, no melting
C sand	sinks, stays visible	no change
D sulphur	floats, not wetted	melts at about 115 C, then burns with blue

(a) Looking at a substance does not tell us how it will behave. Each substance has its own set of properties, and these come from what it is made of. A salt and H sugar both dissolve in water and give clear solutions, but C sand does not dissolve and sinks, and D sulphur floats and is not wetted. E iron filings also do not dissolve, but a magnet pulls them out. In the heat test, H sugar melts at about 186 C and chars, D sulphur melts at about 115 C and burns, but C sand does not change. So powders that look alike have different properties. (b) Physical properties: solubility (salt and sugar dissolve, sand does not) and magnetism (iron is pulled by a magnet); melting point is also acceptable. Chemical property: burning (sulphur burns with a blue flame and a sharp choking smell, but sand does not burn); reacting with acid is also acceptable.

		flame
E iron filings	sink, pulled out by a magnet	glow orange-red, turn dull blackish
H sugar	dissolves, clear solution	melts at about 186 C, browns, then chars

(a) Wanjiru noticed that several of these substances look like powders. Why do substances that look alike behave so differently? Use evidence from at least three substances in the table. (b) Name two physical properties and one chemical property that you used to tell the substances apart.

Section 2: Separating mixture M

Mixture M was 3 g iron filings + 3 g sand + 3 g salt, a total of 9 g, in water. Group 1 separated it in three steps and recorded the masses recovered: iron 2.9 g, sand 2.8 g (after drying), salt 2.7 g.

(a) Name the method that removed each of the three substances and explain why that method worked for that

(a) Iron was removed by magnetic separation, because iron is pulled by a magnet and sand and salt are not. Sand was removed by filtration, because sand does not dissolve, so its grains are trapped on the filter paper while the salt solution passes through. Salt was recovered by evaporation of the filtrate, which leaves the salt behind when the water goes away. (b) Salt dissolves in water, so it is mixed in the water and passes through the filter paper. It is not magnetic, so a magnet cannot pull it out. Only evaporation separates it. (c) Total recovered = $2.9 + 2.8 + 2.7 = 8.4$ g. Loss = $9 - 8.4 = 0.6$ g. The loss was caused by spillage and by some material sticking to the equipment.

substance. (b) Why could the salt not be removed by filtration or a magnet? (c) Calculate the total mass recovered and the mass lost. Give one reason for the loss.	
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Section 3: Classifying and the meaning of Chemistry

(a) Write the meaning of Chemistry and of matter. (b) Classify D sulphur, A table salt and mixture M as an element, a compound or a mixture, and give a reason for each. (c) Name the five branches of chemistry and say which branch you would link to the separation of mixture M.	(a) Chemistry is the science that studies matter: its composition, properties, changes and uses. Matter is anything that has mass and occupies space. (b) D sulphur is an element, like E iron. A table salt is a compound because it is a substance made of more than one kind of element joined together (sodium chloride), like B water and H sugar. Mixture M is a mixture because it contains iron filings, sand and salt that are not joined together and could be separated by magnet, filtration and evaporation. (c) The five branches are organic, inorganic, physical, analytical and biochemistry. The separation of mixture M links to analytical chemistry.
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Section 4: Medicines and using them safely

Wanjiru asked why the antacid and paracetamol work differently. (a) Using what you saw in the vinegar test, explain how F antacid tablet relieves heartburn and how G paracetamol tablet is different. (b) The adult paracetamol label says 1 to 2 tablets of 500 mg every 4 to 6 hours. Calculate the maximum number of milligrams and grams allowed in 24 hours if the maximum is 8 tablets, and say what can happen if more is taken. (c) Otieno's customer wanted to give an adult dose to a small child. What advice would you give, and which items on a label help a buyer to be safe?	(a) F antacid fizzes when dropped into dilute vinegar, which is an acid, so it reacts with and neutralises acid. Heartburn is caused by too much acid in the stomach, so the antacid relieves it by neutralising that acid. G paracetamol does not fizz in vinegar, so it does not neutralise acid. It relieves pain and fever by a different action. (b) $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ in 24 hours. Taking more than this can damage the liver. (c) Do not give a child an adult dose. Read the label, follow the correct dosage and ask a health worker or pharmacist for advice. Label items that help a buyer are the quality mark (such as the KEBS Diamond Mark), the expiry date, the ingredients and preservatives, nutritional information and caution messages. Too much antacid taken too often can also cause constipation or diarrhoea, so both medicines are harmful in excess. Avoiding substance abuse (using a substance in a harmful way or amount) keeps people safe.
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Section 5: Final Explanation

Write a connected explanation of 8-10 sentences that answers the Driving Question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? Use evidence from the Kisumu tray, including at least one number from your results. Include (i) why substances that look alike behave differently, (ii) how chemists use properties to separate or make materials, with at least one Kenyan industry or example, (iii) why the antacid and paracetamol work differently and why both are harmful in excess, and (iv) one responsibility you will take for a safe and healthy environment at school and at home.

Each substance on the Kisumu tray has its own set of properties, such as solubility, magnetism, melting point and burning, and these come from what the substance is made of. That is why powders that look alike behave differently: A salt and H sugar dissolve and cannot be filtered out, but C sand and E iron filings do not dissolve, and D sulphur burns with a blue flame while sand does not. Chemists use these properties to choose, separate and make materials. For example, in mixture M we used a magnet to remove iron (2.9 g), filtration to remove sand (2.8 g) and evaporation to recover salt (2.7 g), which shows that the method must match the property. Kenyan industries do the same: Magadi soda ash and Mumias and Nzoia sugar depend on separating substances, and pharmaceutical manufacturers in Nairobi choose materials by their properties to make medicines. The antacid F neutralises excess stomach acid because it reacts with acid, so it relieves heartburn, while paracetamol G relieves pain and fever by a different action and does not neutralise acid. Both are harmful in excess: adults must not take more than $8 \times 500 \text{ mg} = 4 \text{ g}$ of paracetamol in 24 hours because it can damage the liver, and too much antacid can cause constipation or diarrhoea. Labels with a quality mark, expiry date, ingredients and caution messages, correct dosage, a health worker's advice and avoiding substance abuse turn this knowledge into safe use. Children must never be given adult doses. At school I will take responsibility by wearing goggles, not tasting substances and following my teacher's safety rules, and at home I will read labels and keep medicines away from small children.

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the Kisumu tray	Uses accurate results for several substances A-H, including exact numbers and units (such as 2.9 g, 2.8 g, 2.7 g, 8.4 g, 0.6 g, 4 g), and links each piece of evidence to a claim.	Uses correct evidence from some substances and at least one correct number, but links to the claim are sometimes missing.	Gives few or inaccurate results, or makes claims without evidence from the tray.
Explaining properties and separation	Clearly explains that properties come from what a substance is made of, and correctly matches magnetic separation, filtration and evaporation to the properties of iron, sand and salt, including why dissolved salt needs evaporation.	Names the methods and most properties correctly, with a partly complete reason for why each method works.	Confuses methods or properties, or only states that substances are different without a reason.
Chemistry concepts and	Correctly defines chemistry and matter,	Gives mostly correct definitions and	Definitions or classifications are

classification	classifies elements, compounds and mixtures with reasons, and names the five branches with a correct link.	classifications, with a reason missing in places.	incomplete or mostly incorrect.
Medicines and safe use	Explains that the antacid neutralises acid and paracetamol acts differently, calculates $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$ correctly, explains the harm of excess, and links label items, correct dosage and avoiding substance abuse to safety.	Explains the medicines and calculation correctly but gives limited detail on label items or harm.	Mixes up the two medicines, makes a calculation error, or gives little on safety.
Final explanation: clarity, Kenyan context and responsibility	Gives a connected 8-10 sentence answer to the driving question, uses a Kenyan industry or example, and states a specific, realistic responsibility for school and home.	Answer is mostly connected and answers the driving question, with a general Kenyan example or responsibility.	Answer is a list of unconnected facts, is too short, or does not address the driving question.

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